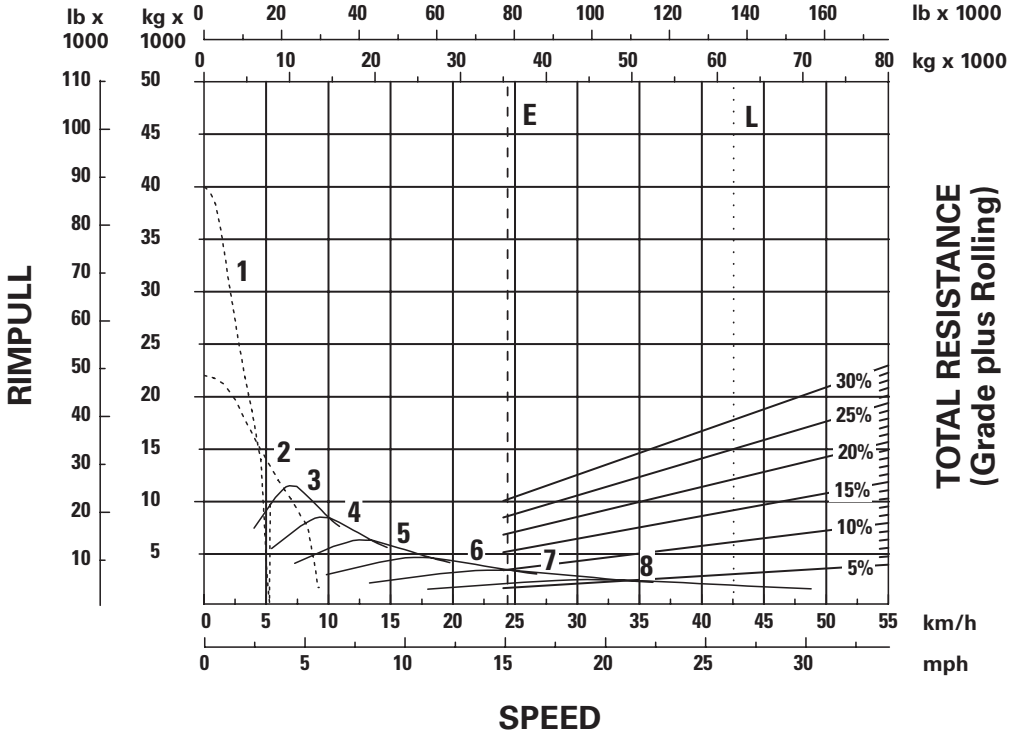
STANDARD ARRANGEMENT*
GROSS WEIGHT



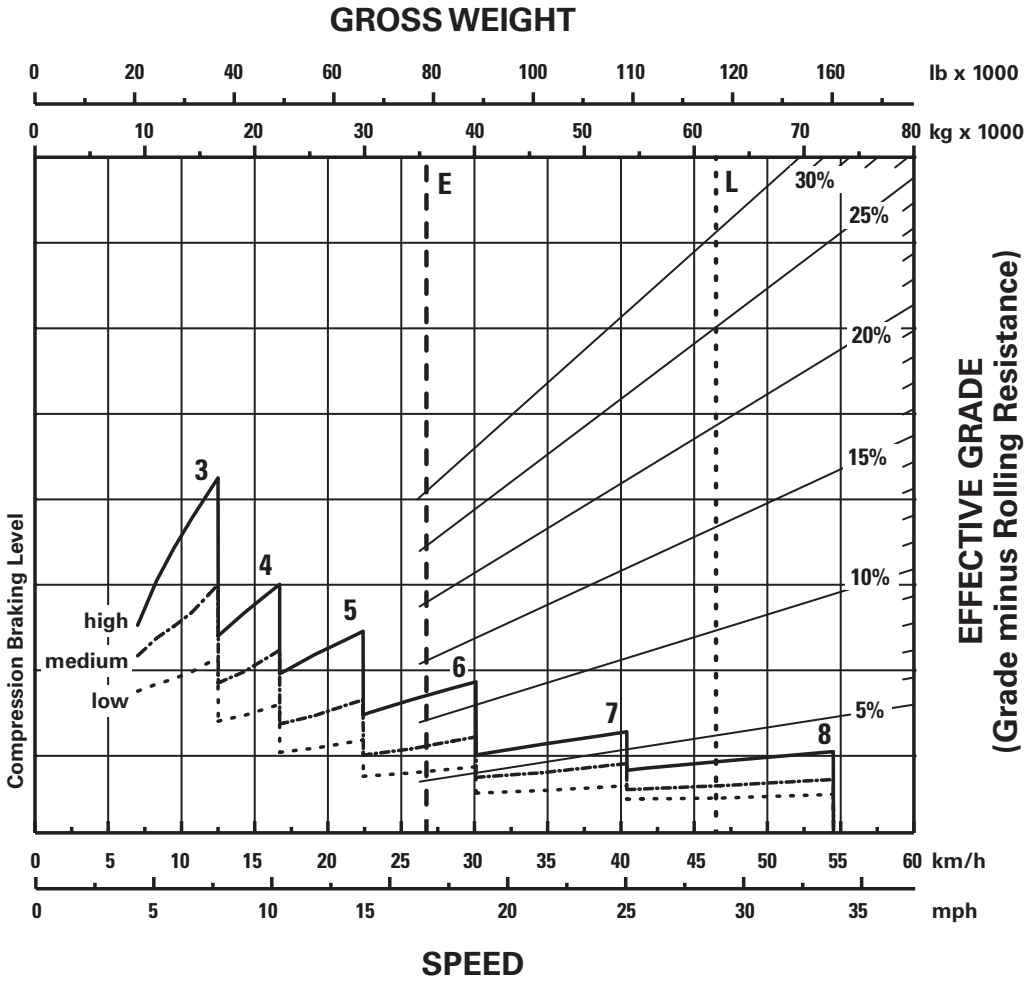
*at sea level

KEY

1 — 1st Gear Torque Converter Drive
2 — 2nd Gear Torque Converter Drive
3 — 3rd Gear Direct Drive
4 — 4th Gear Direct Drive
5 — 5th Gear Direct Drive
6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 35 808 kg (78,943 lb)
L — Loaded 61 935 kg (136,553 lb)



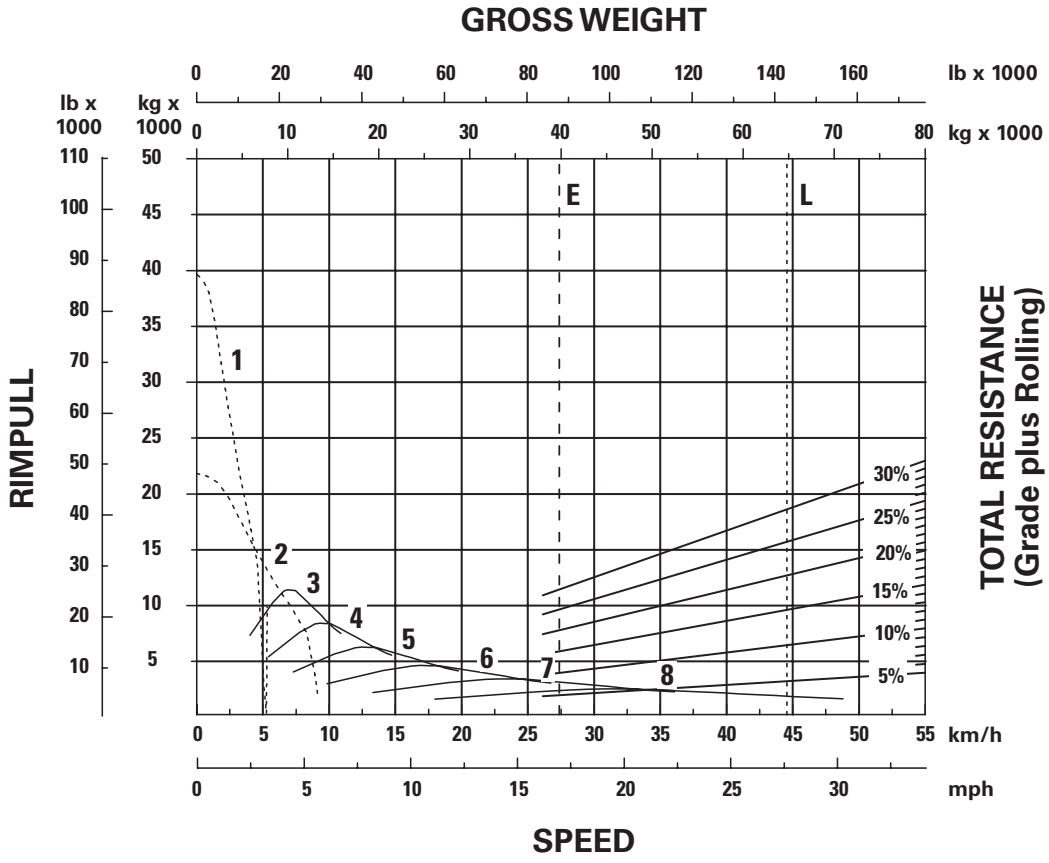
*at sea level

KEY

3 — 3rd Gear Direct Drive
4 — 4th Gear Direct Drive
5 — 5th Gear Direct Drive
6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 35 808 kg (78,943 lb)
L — Loaded 61 935 kg (136,553 lb)

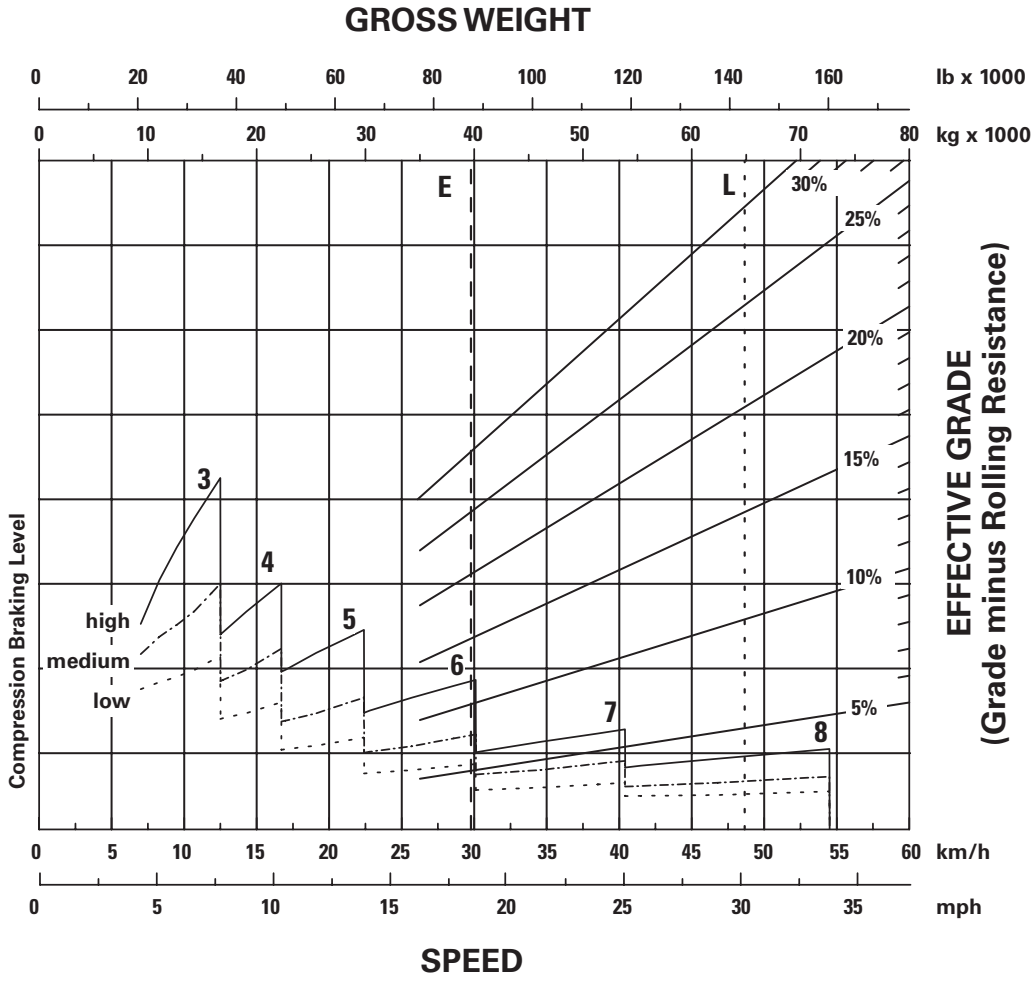


KEY

1 — 1st Gear Torque Converter Drive
2 — 2nd Gear Torque Converter Drive
3 — 3rd Gear Direct Drive
4 — 4th Gear Direct Drive
5 — 5th Gear Direct Drive
6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 39 866 kg (87,809 lb)
L — Loaded 64 904 kg (143,009 lb)



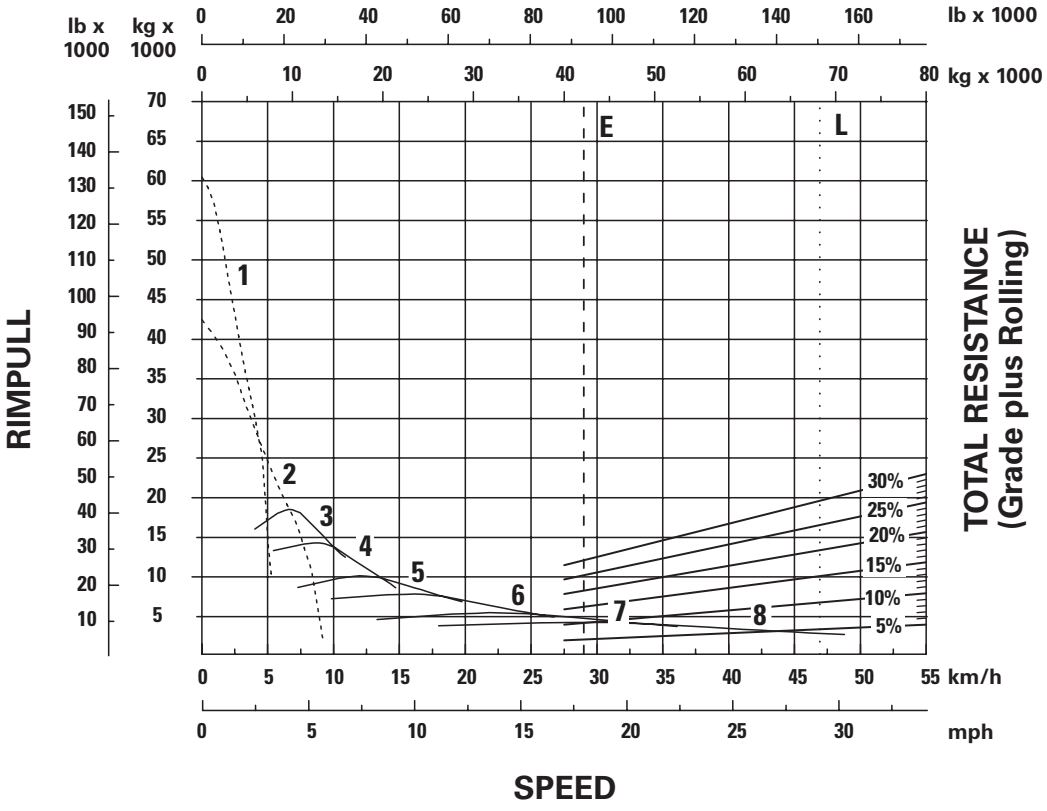
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6 — 6th Gear Direct Drive
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L — Loaded 64 904 kg (143,009 lb)

STANDARD ARRANGEMENT*
GROSS WEIGHT



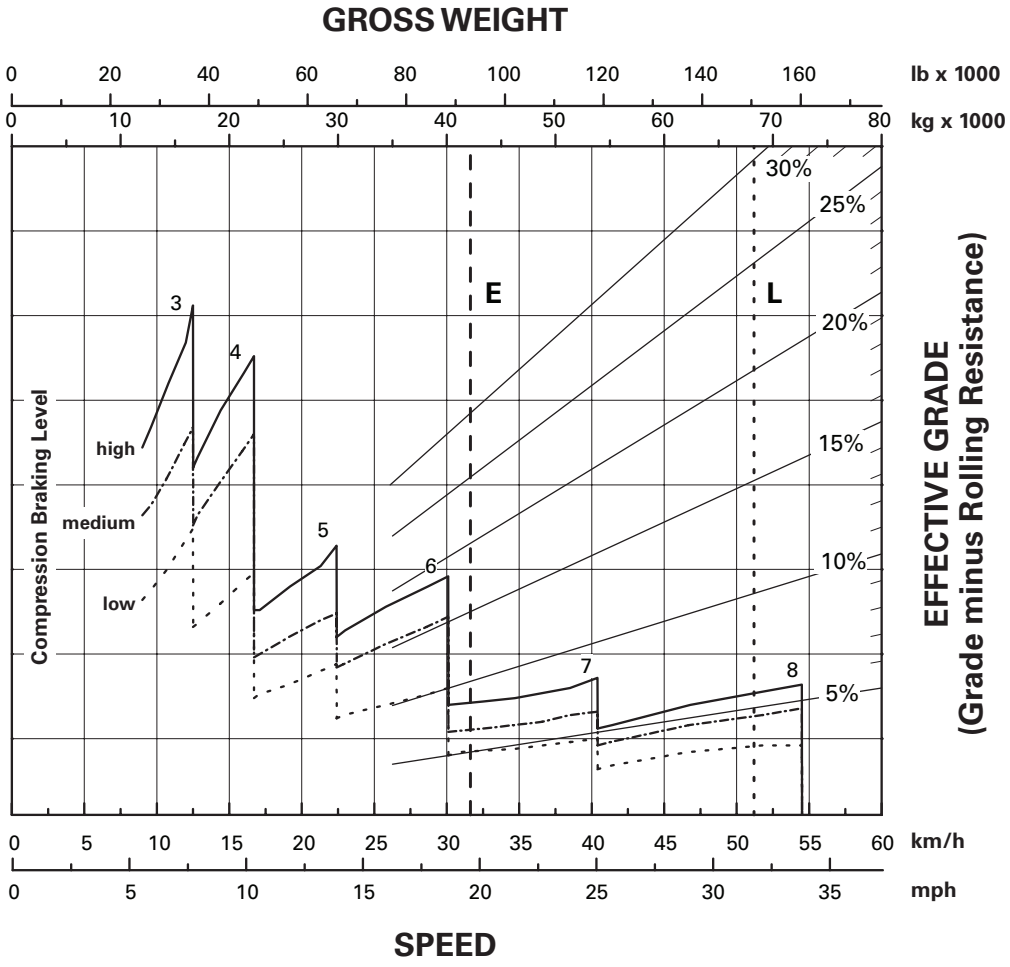
*at sea level

KEY

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3 — 3rd Gear Direct Drive
4 — 4th Gear Direct Drive
5 — 5th Gear Direct Drive
6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 42 158 kg (92,942 lb)
L — Loaded 68 289 kg (150,552 lb)



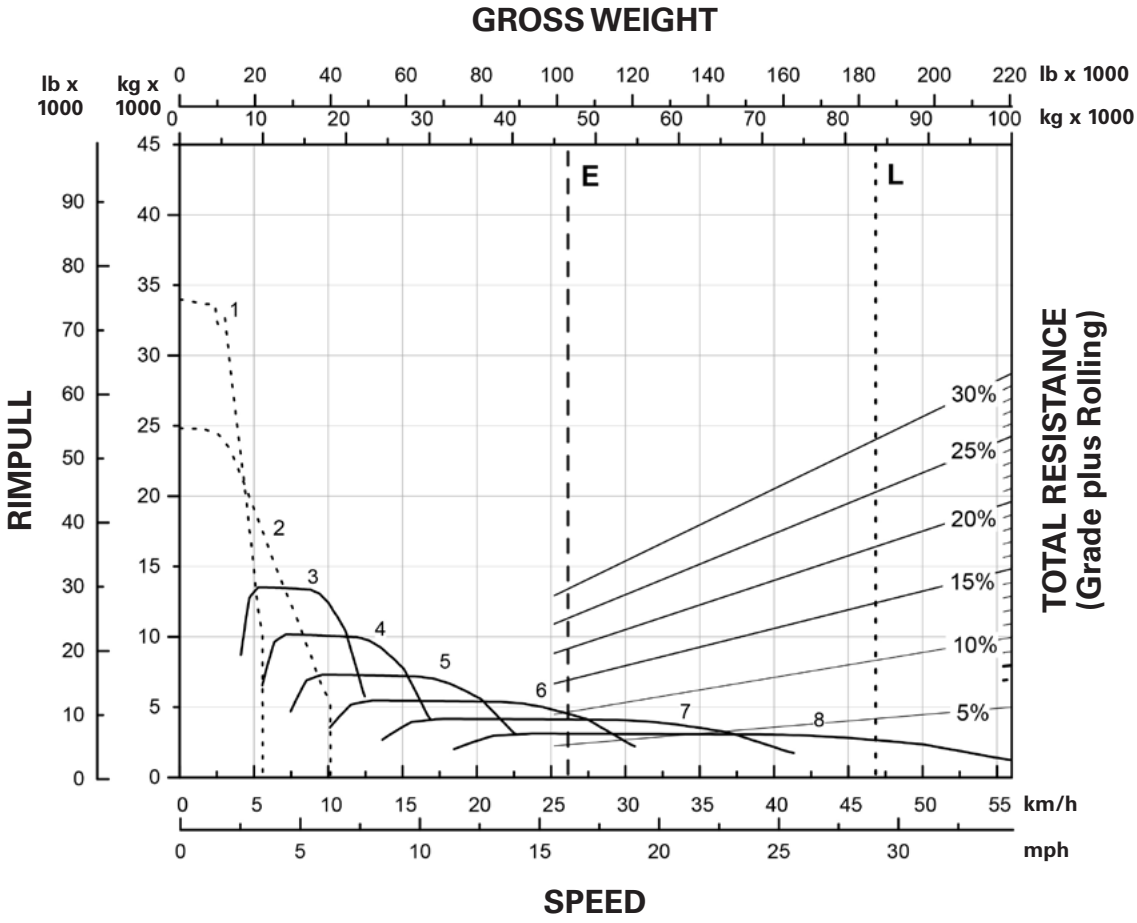
*at sea level

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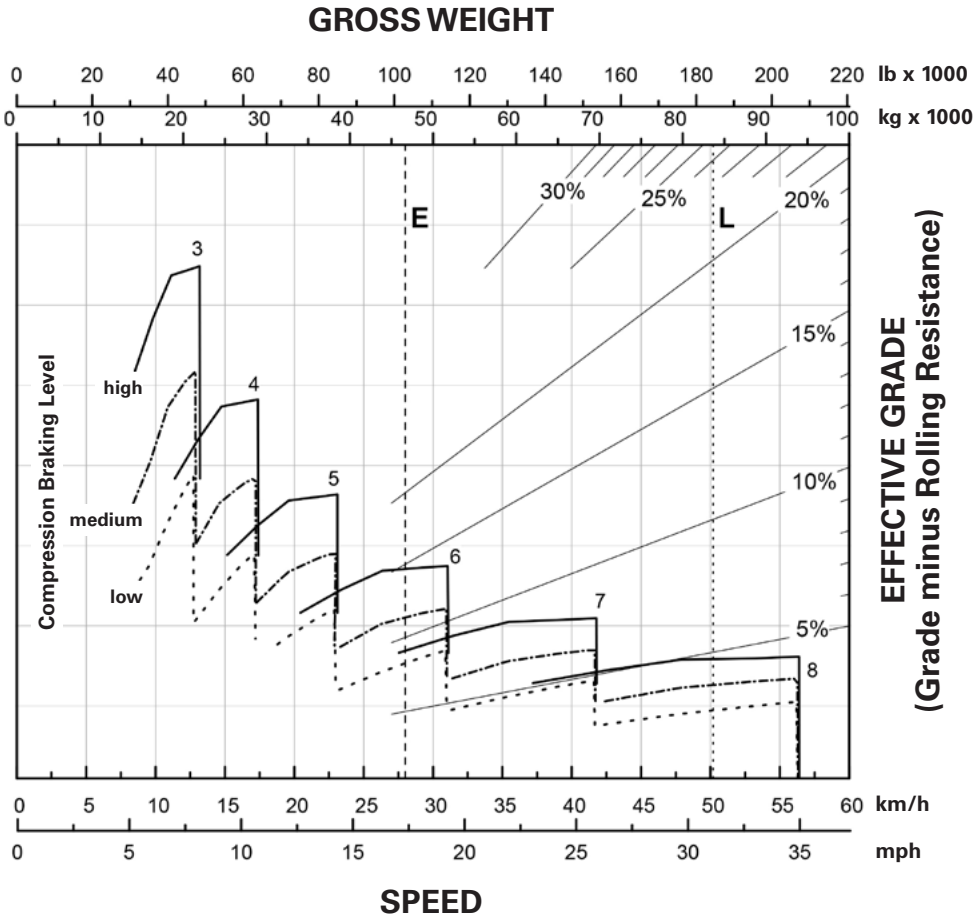


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6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 46 607 kg (102,750 lb)
L — Loaded 83 892 kg (184,950 lb)

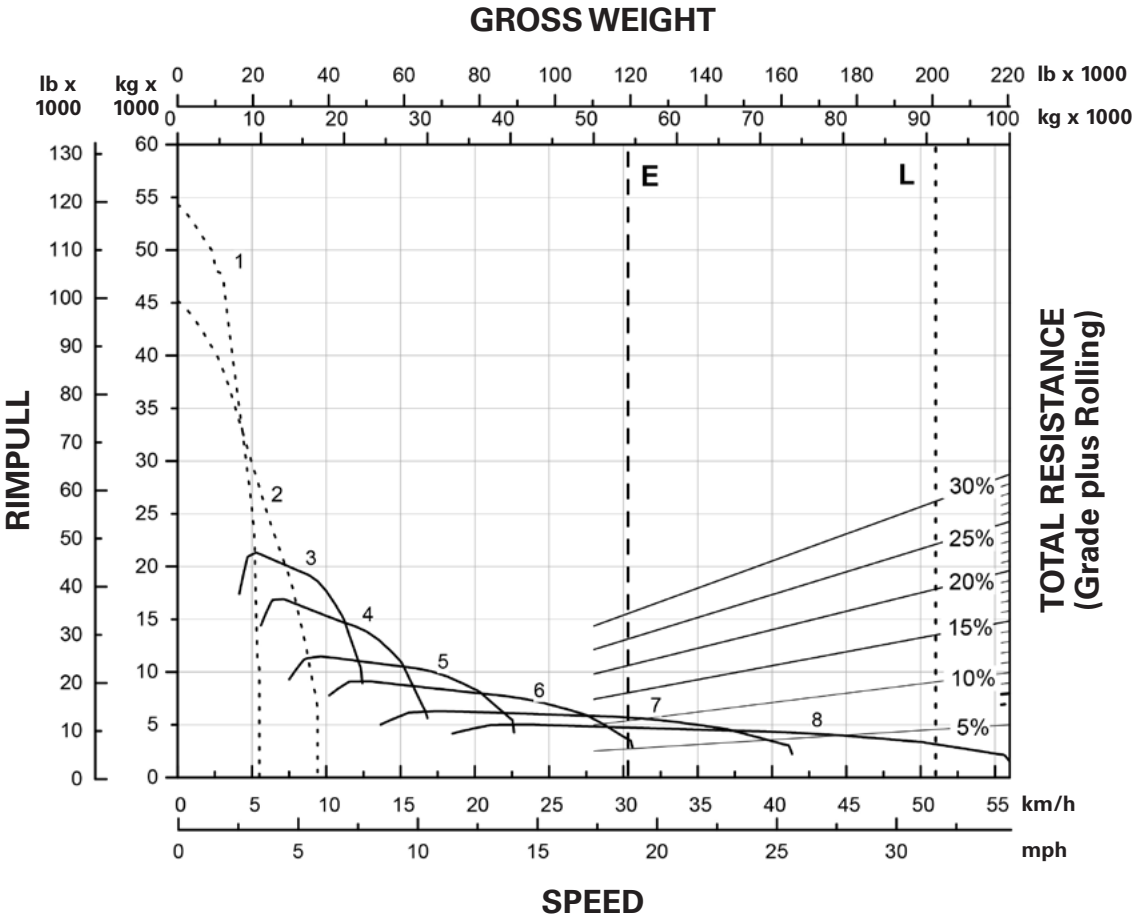


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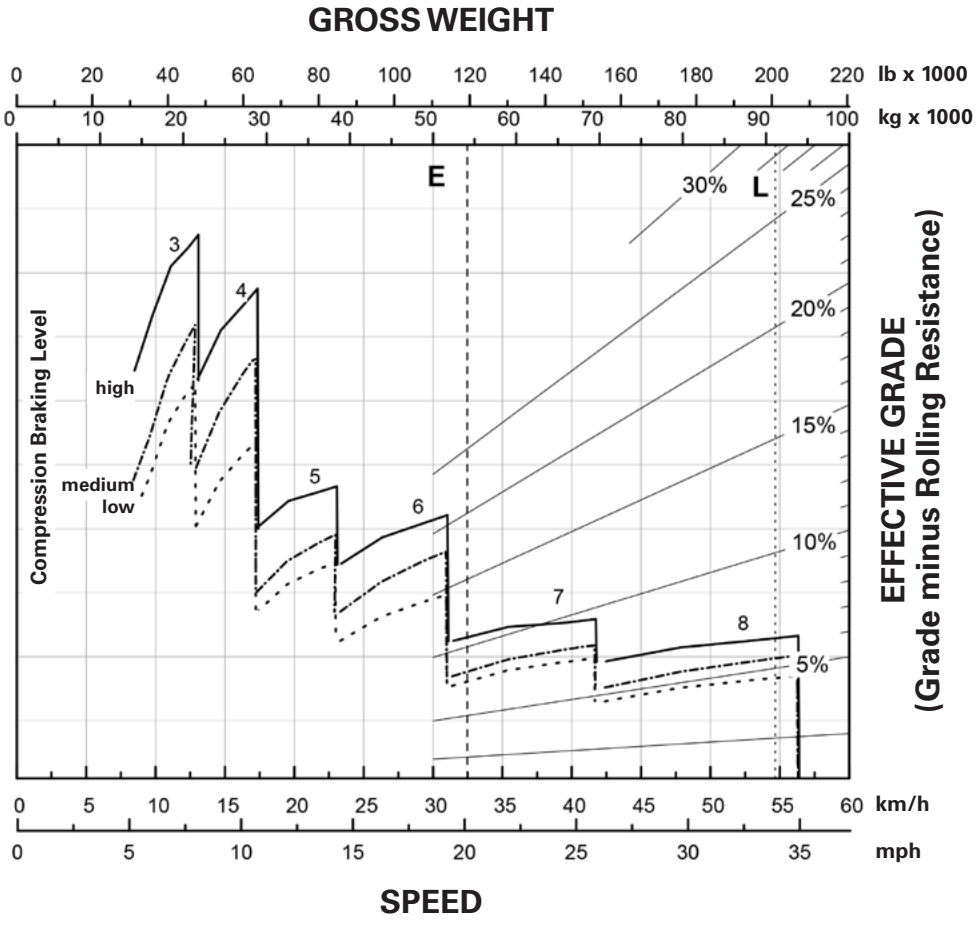


KEY

1 — 1st Gear Torque Converter Drive
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3 — 3rd Gear Direct Drive
4 — 4th Gear Direct Drive
5 — 5th Gear Direct Drive
6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 54 005 kg (119,060 lb)
L — Loaded 91 290 kg (201,260 lb)

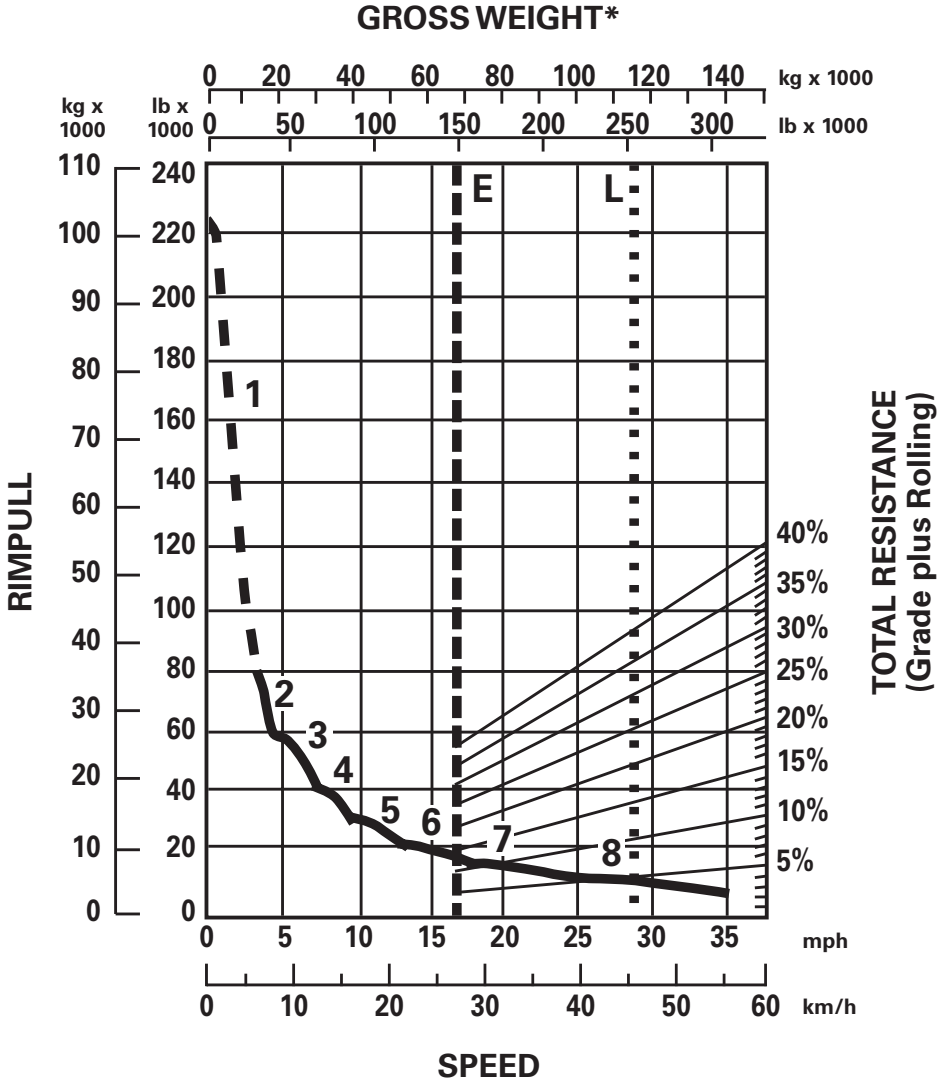


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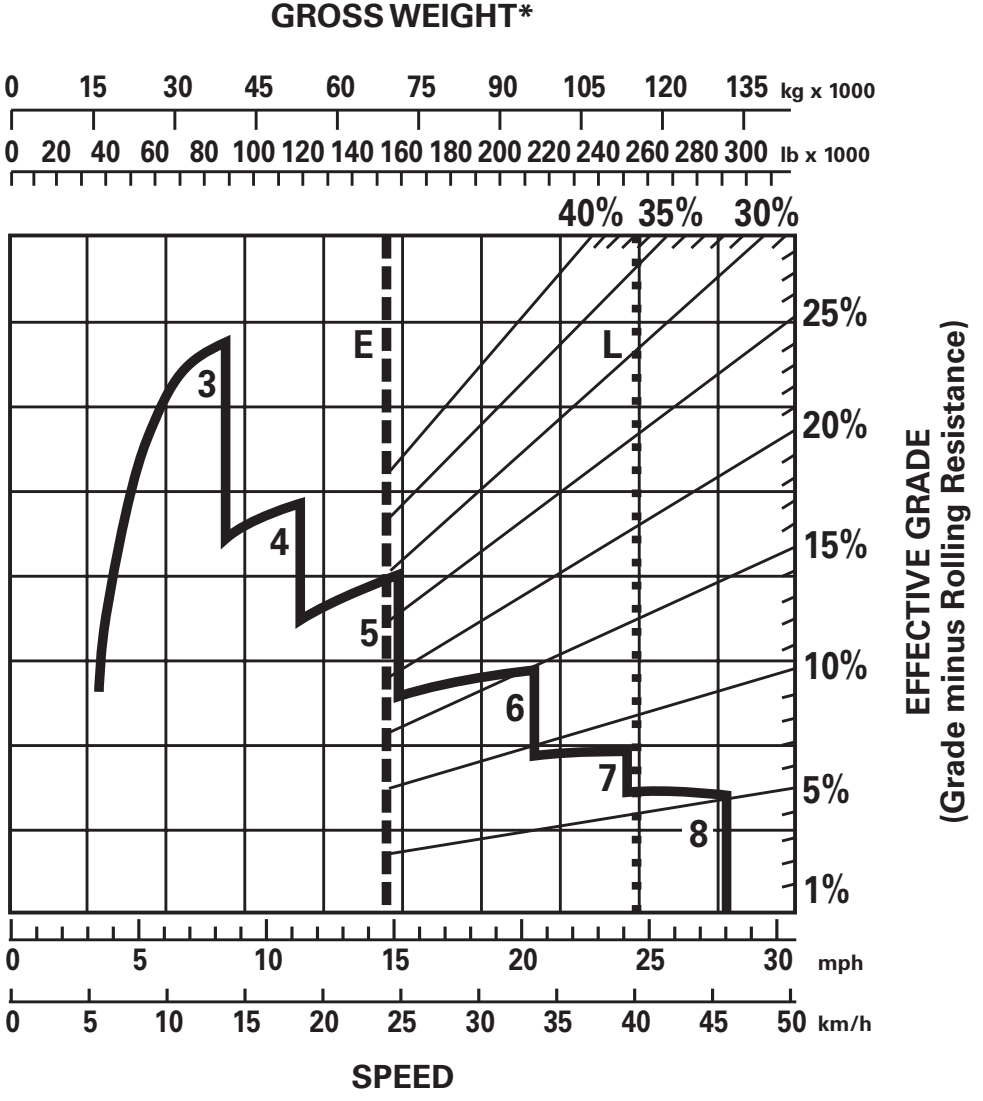
*at sea level

KEY

1 — 1st Gear Torque Converter Drive
2 — 2nd Gear Torque Converter Drive
3 — 3rd Gear Direct Drive
4 — 4th Gear Direct Drive
5 — 5th Gear Direct Drive
6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 72 804 kg (160,505 lb)
L — Loaded 119 978 kg (264,505 lb)



*at sea level

KEY

3 — 3rd Gear Direct Drive
4 — 4th Gear Direct Drive
5 — 5th Gear Direct Drive
6 — 6th Gear Direct Drive
7 — 7th Gear Direct Drive
8 — 8th Gear Direct Drive

KEY

E — Empty 72 804 kg (160,505 lb)
L — Loaded 119 978 kg (264,505 lb)

ESTIMATING OWNING & OPERATING COSTS

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General

Machine users must balance productivity and costs to achieve optimum performance ... that is, achieve the desired production at the lowest possible cost. The approach most often used to measure machine performance is this simple equation:

$$\frac{\textit{Lowest Possible Hourly Costs}}{\textit{Highest Possible Hourly Productivity}} = \textit{Top Machine Performance}$$

Most sections of this Handbook deal with the productivity of Cat machines. This section considers the cost aspect of performance.

Hourly Owning and Operating Costs for a given machine can vary widely because they are influenced by many factors: the type of work the machine does, the ownership period, local prices of fuel and labor, the repair and maintenance costs, shipping costs from the factory, interest rates, etc. No attempt is made in this handbook to provide precise hourly costs for each model. Users must be able to estimate with a reasonable degree of accuracy what a machine will cost per hour to own and operate in a given application and locality. Therefore, this section provides a suggested method of estimating hourly owning and operating costs. When this method is coupled with local conditions and dealer input, it will result in reasonable estimates.

The method suggested follows several basic principles:

- Repair and Planned Maintenance cost per hour are developed jointly by the customer and local Cat dealer.
- In the examples, labor is assumed @ $60.00 per hour, fuel @ $1.25 per gallon. For reliable estimates, these costs must always be obtained locally.
- Because of different standards of comparison, what may seem a high application to one machine owner may appear only medium to another.
- Unless otherwise specified, the word "hour" when used in this section means clock or operating hours, not Service Meter Units.

NOTE: An Excel O&O Estimating Form is available, in electronic spreadsheet format, to calculate an Hourly Owning & Operating Cost Estimate based on the procedure outlined in this section.
The spreadsheet my be accessed via the website at https://dealer.cat.com – From the home page, select "Product Support," "Equipment Management Solutions," "Owning & Operating Costs." Under the heading "Machine Technical Information," select "O&O COST ESTIMATING Form."

HOURLY OWNING AND OPERATING COST ESTIMATE

DATE _____

	Estimate #1	Estimate #2
A–Machine Designation	_____	_____
B–Estimated Ownership Period (Years)	_____	_____
C–Estimated Usage (Hours/Year)	_____	_____
D–Ownership Usage (Total Hours)(B × C)	_____	_____

OWNING COSTS

1. a. Delivered Price (P), to the Customer (including attachments)	_____	_____
b. Less Tire Replacement Cost if desired	_____	_____
c. Delivered Price Less Tires	_____	_____
2. Less Residual Value at Replacement (S) (See subsection 2A on back)	(___%) _____	(___%) _____
3. a. Net Value to be recovered through work (line 1c less line 2)	_____	_____
b. Cost Per Hour: $\frac{\text{Net Value}}{\text{Total Hours}}$ (1) _____ (2) _____	_____	_____

4. Interest Costs N = No. Yrs. $\dfrac{\dfrac{P(N+1)+S(N-1)}{2N} \times \text{Simple Int. \% Rate}}{\text{Hours/Year}} =$

(1) $\frac{+1}{\quad} + \frac{-1}{\quad} \times$ ___ % (2) $\frac{+1}{\quad} + \frac{-1}{\quad} \times$ ___ %

_____ = _____ = _____ _____

_____ Hours/Yr. _____ Hours/Yr.

5. Insurance N = No. Yrs. $\dfrac{\dfrac{P(N+1)+S(N-1)}{2N} \times \text{Insurance \% Rate}}{\text{Hours/Year}} =$

(1) $\frac{+1}{\quad} + \frac{-1}{\quad} \times$ ___ % (2) $\frac{+1}{\quad} + \frac{-1}{\quad} \times$ ___ %

_____ = _____ = _____ _____

_____ Hours/Yr. _____ Hours/Yr.

(Optional method when Insurance cost per year is known)

Ins. $ _____ Per Yr. ÷ _____ Hours/Yr. =

	Estimate #1	Estimate #2
6. Property Tax $\frac{P(N+1)+S(N-1)}{2N} \times$ Tax Rate % / Hours/Year =		
(1) $\frac{+1}{\quad}$ + $\frac{-1}{\quad}$ × ___ % (2) $\frac{+1}{\quad}$ + $\frac{-1}{\quad}$ × ___ %		
_____ = _____ Hours/Yr. _____ = _____ Hours/Yr.		
(Optional method when Property Tax cost per year is known)		
Property Tax $ _____ Per Yr. ÷ _____ Hours/Yr. =		
7. TOTAL HOURLY OWNING COST (add lines 3b, 4, 5 and 6)		
OPERATING COSTS		
8. Fuel: Unit Price × Consumption		
(1) _____ × _____ =		
(2) _____ × _____ =		
9. Planned Maintenance (PM)-Lube Oils, Filters, Grease, Labor: (contact your local Cat dealer).		
10. a. Tires: Replacement Cost ÷ Life in Hours		
$\frac{\text{Cost}}{\text{Life}}$ (1) _____ (2) _____		
b. Undercarriage (Impact + Abrasiveness + Z Factor) × Basic Factor		
(1) (_____ + _____ + _____) = _____ × _____ =		
(2) (_____ + _____ + _____) = _____ (Total) × _____ (Factor) =		
11. Repair Cost (Per Hour) (contact your local Cat dealer).		
12. Special Wear Items: Cost ÷ Life . . . . . (See subsection 12A on back)		
13. TOTAL OPERATING COSTS (add lines 8, 9, 10a (or 10b), 11 and 12).		
14. MACHINE OWNING PLUS OPERATING (add lines 7 and 13)		
15. OPERATOR'S HOURLY WAGE (include fringes)		
16. TOTAL OWNING AND OPERATING COST		

SUBSECTION 2A: Residual Value at Replacement

Gross Selling Price	(est. #1) (___%)	_____	(est. #2) (___%)	_____	
Less: a. Commission		_____		_____	
b. Make-ready costs		_____		_____	
c. Inflation during ownership period*		_____		_____	
Net Residual Value (Enter on line 2)		_____	(___%)	_____	(___%) of original delivered price

*When used equipment auction prices are used to estimate residual value, the effect of inflation during the ownership period should be removed to show in constant value what part of the asset must be recovered through work.

SUBSECTION 12A: Special Items
(cutting edges, ground engaging tools, bucket teeth, etc.)

(1) Cost		Life		Cost/Hour	(2)				
1. _____	÷	_____	=	_____	1. _____	÷	_____	=	_____
2. _____	÷	_____	=	_____	2. _____	÷	_____	=	_____
3. _____	÷	_____	=	_____	3. _____	÷	_____	=	_____
4. _____	÷	_____	=	_____	4. _____	÷	_____	=	_____
5. _____	÷	_____	=	_____	5. _____	÷	_____	=	_____
6. _____	÷	_____	=	_____	6. _____	÷	_____	=	_____
		Total	(1)	_____	(2)		_____		

(Enter total on line 12)

1-7 ESTIMATING OWNING COSTS

(Line Items 1 through 7)

To protect their equipment investment and be able to replace it, the machine owner must recover over the ownership period an amount equal to the loss in resale value plus the other costs of owning the equipment including interest, insurance and taxes.

The machine owner, for accounting purposes, estimates resale value loss in advance, and recovers his original equipment investment by establishing depreciation schedules according to the various uses of the equipment. Proper financial and tax assistance is highly recommended when establishing depreciation schedules.

Considering today's economic conditions worldwide and the trend toward larger, more expensive equipment, many users choose to keep these units on the job well after they have been fully depreciated for tax purposes. On the other hand, tax incentives in many areas may favor trading a machine well before that occurs.

The ownership period in years, the hours per year, and the total number of hours on a machine, are significant factors in determining O&O costs. Additionally, since the ownership period and machine hours can vary widely for different customers for a given model, it is not practical to calculate O&O costs using an assumed ownership period. The customer must provide that information for each situation.

These same factors will be used to develop the Repair costs and Planned Maintenance costs by your local Cat dealer.

The machine depreciation method suggested in this handbook is not based on or related to any tax considerations, but rather is a simple straight line write-off based solely on the number of years and hours the owner expects to use during the ownership period.

Accordingly, it is imperative that careful consideration be given the selection of depreciation periods, and that for owning and operating cost calculations they be based on actual ownership periods and hours on the machine rather than tax write-off life.

Typical Application Descriptions

The following tables show typical descriptions for work performed by each product family for three different application levels. It is only a guide and can be used along with the fuel and tire charts to help determine fuel and tire cost factors. Additionally, many times the ownership period and the number of hours per year a machine is used, is related to application.

1 DELIVERED PRICE TO CUSTOMER

(Line Item 1a, b and c)

Delivered price should include all costs of putting a machine on the user's job including transportation and any applicable sales taxes.

On rubber tired machines, tires are considered a wear item and covered as an operating expense. Accordingly, some users may wish to deduct tire costs from the delivered price particularly for larger machines.

2 RESIDUAL VALUE AT REPLACEMENT

(Line Item 2 and Subsection 2A)

Any piece of earthmoving machinery will have some residual value at trade-in. While many owners prefer to depreciate their equipment to zero value, others recognize the residual resale or trade-in value. This is at the estimator's option, but as in the discussion of depreciation, today's higher equipment costs almost dictate that resale value be considered in determining the net depreciable investment. And if machines are traded early for tax incentive purposes, resale value becomes even more significant.

For many owners, potential resale or trade-in value is a key factor in their purchasing decisions, since this is a means of reducing the investment they must recover through depreciation charges. The high resale value of Caterpillar built machines can reduce hourly depreciation charges, lower total hourly owning costs and improve the owner's competitive position.

When resale or trade-in value is used in estimating hourly owning and operating costs, local conditions must be considered, as used equipment values vary widely around the world. However, in any given used equipment business, factors which have greatest influence on resale or trade-in value are: age of machine (years), the number of hours on the machine at the time of sale or trade, the type of jobs and operating conditions in which it worked, and the physical condition of the machine. Your local Cat dealer is your best source for determining current used equipment values.

Subsection 2A can be used to calculate the estimated residual value. If recent auction prices for used machines are used as a guide, then the value (or percentage) should be adjusted downward to remove the effect of inflation. Governmental indices on construction equipment costs or Dealer price records can be used to calculate the amount of inflation for the appropriate useful life. Another way to estimate residual value is comparing the current used machine value to the current new machine price provided major product changes haven't occurred.

3 VALUE TO BE RECOVERED THROUGH WORK

(Line Item 3a and b)

The delivered price (P) less the estimated residual value (S) results in the value to be recovered through work, divided by the total usage hours, gives the hourly cost to protect the asset's value.

4 INTEREST

(Line Item 4)

Many owners charge interest as part of hourly owning and operating costs, others consider it as general overhead in their overall operation. When charged to specific machines, interest is usually based on the owner's average annual investment in the unit.

Interest is considered to be the cost of using capital. The interest on capital used to purchase a machine must be considered, whether the machine is purchased outright or financed.

If the machine will be used for N years (where N is the number of years of use), calculate the average annual investment during the use period and apply the interest rate and expected annual usage:

$$\frac{\left[\frac{P(N+1)+S(N-1)}{2N}\right] \times \text{Simple Int. \% Rate}}{\text{Hours/Year}}$$

5-6 INSURANCE AND TAXES

(Line Items 5 and 6)

Insurance cost and property taxes can be calculated in one of two ways. If the specific annual cost is known, this figure should be divided by the estimated usage (hours/years) and used. However, when the specific interest and tax costs for each machine are not known, the following formulas can be applied:

$$\frac{\text{Insurance}}{\text{N = No. Years}}$$

$$\frac{\left[\frac{P(N+1)+S(N-1)}{2N}\right] \times \text{Insurance \% Rate}}{\text{Hours/Year}}$$

$$\frac{\text{Property Tax}}{\text{N = No. Years}}$$

$$\frac{\left[\frac{P(N+1)+S(N-1)}{2N}\right] \times \text{Tax Rate \%}}{\text{Hours/Year}}$$

8-13 ESTIMATING OPERATING COSTS

(Line Items 8 through 13)

8 FUEL CONSUMPTION

(Line Item 8)

Actual fuel consumption should be measured in the field. However, if no opportunity exists to do this, fuel consumption estimates can be provided by the Caterpillar dealer based on Product Link™ data and experience in the application.

To estimate hourly fuel cost, determine the hourly fuel consumption and multiply by the local unit price of fuel.

Hourly consumption × Local Unit Price of Fuel = Hourly Fuel Cost

9 PLANNED MAINTENANCE (PM) LUBE OILS, FILTERS, GREASE, LABOR

Planned Maintenance (PM) costs should be developed by the Cat dealer, with customer input for the specific application.

PM costs include the parts and labor at the intervals specified in the Operation and Maintenance Manuals provided for each machine. PM costs for each machine may vary slightly depending upon factors required or specified by the customer. See your local Cat dealer to develop the specific PM cost per hour estimate for your machine and application.

10a TIRES

(Line Item 10a)

Tire costs are an important part of the hourly cost of any wheel machine. Tire costs are also one of the most difficult to predict with many variables. The best estimate for tire costs are obtained when tire life estimates are based upon actual customer experience, and are used with prices the machine owner actually pays for the replacement tires.

For cases in which tire experience is not available, use the following tire life estimator curves.

Tire Life Estimators

- Curves do not allow for additional life from recapping. They assume new tires run to destruction, but this is not necessarily recommended.
- Based on standard machine tires. Optional tires will shift these curves either up or down.
- Sudden failure (blow out) due to exceeding Ton-MPH (tkm/h) limitations is not considered. Nor are premature failures due to puncture.
- Application Descriptions/Zones:

  Low/Zone A: almost all tires actually wear through the tread from abrasion.

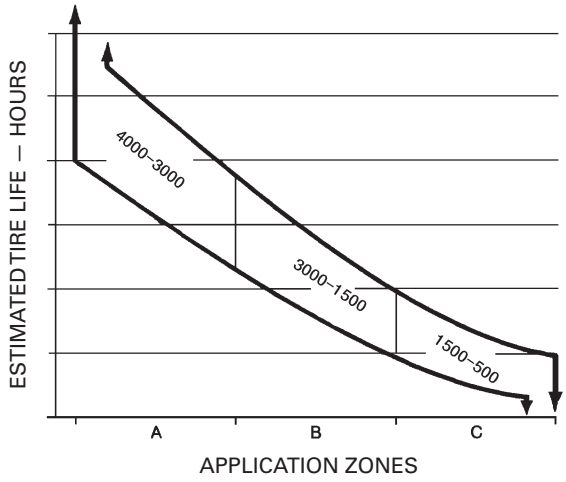
  Medium/Zone B: tires wear out normally but others fail prematurely due to rock cuts, impacts and non-repairable punctures.

  High/Zone C: few, if any, tires wear through the tread due to non-repairable damages, usually from rock cuts, impacts and continuous overloading.

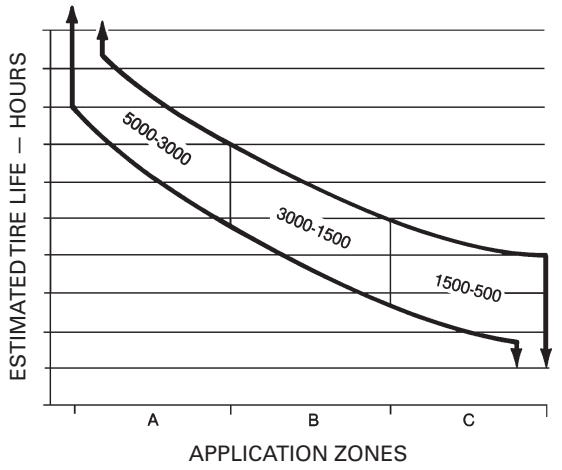
NOTE: Tire life can often be increased by using extra tread and extra deep tread tires.

NOTE: Premature failure could occur at any time due to puncture.

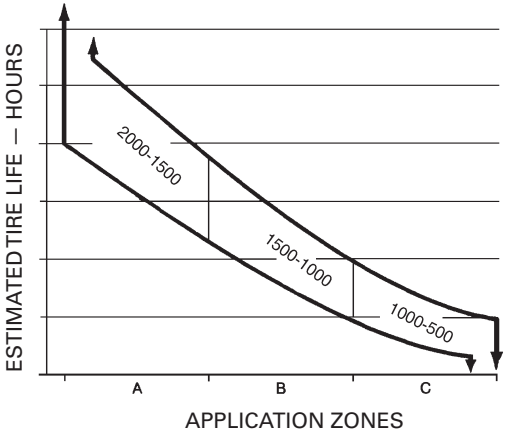
MOTOR GRADERS



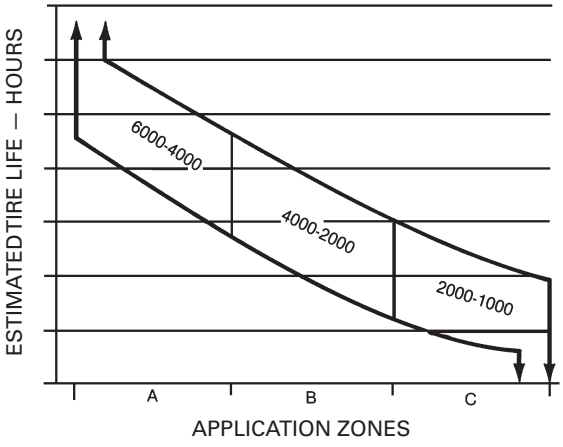
WHEEL TRACTOR-SCRAPERS



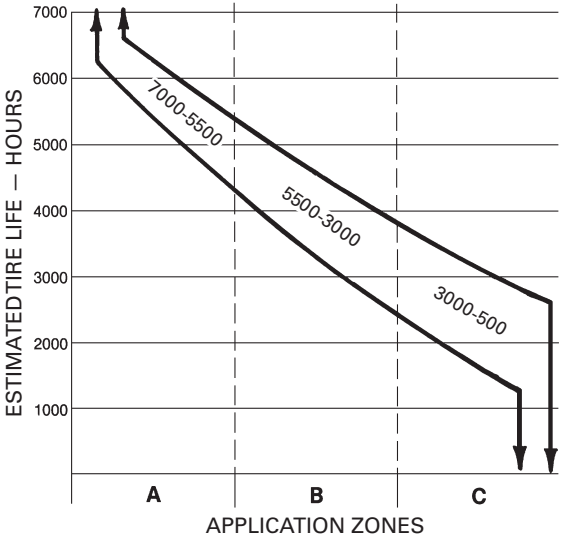
LOAD HAUL DUMP UNITS



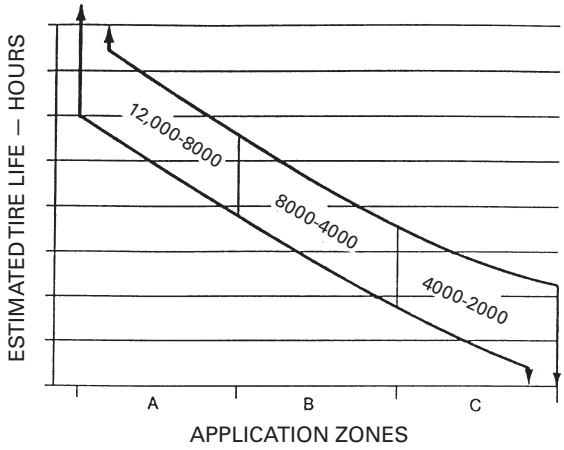
MINING & OFF-HIGHWAY TRUCKS



ARTICULATED TRUCKS



TRACTORS/WAGONS



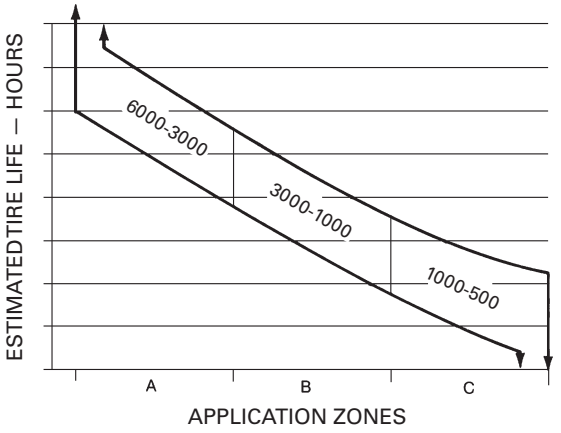
Key:

Zone A — Almost all tires actually wear through the tread due to abrasion.

Zone B — Some tires wear out normally while others fail prematurely due to rock cuts, impacts and non-repairable punctures.

Zone C — Few, if any, tires wear through the tread because of non-repairable damages, usually from rock cuts, impacts or continuous overloading.

WHEEL TRACTORS
WHEEL LOADERS



Key:
Zone A — Almost all tires actually wear through the tread due to abrasion.
Zone B — Some tires wear out normally, others fail prematurely due to rock cuts, impacts, and non-repairable punctures.
Zone C — Few, if any, tires wear through the tread because of non-repairable damages, usually from rock cuts, impacts, or continuous overloading.

GOODYEAR LIFE ESTIMATING SYSTEM

As an additional assist in estimating *hauling unit* tire life, Goodyear Tire and Rubber Co. has furnished the following information which is included here with their permission. READ THE PREAMBLE CAREFULLY.

"… at present, there is no completely accurate, foolproof method of forecasting tire life. Tire engineers have many theoretical methods … but these generally are so involved and time consuming that they are impractical for field use.

"However, the tire industry has made many surveys of tire performance and arrived at a system which can give rough *estimates* of tire life. Studies done by the major tire companies and by at least two major equipment manufacturers are in close agreement.

"The table [which follows] shows how to apply this system …"

ESTIMATED TIRE SERVICE LIFE OF HAULING UNITS (Trucks and Scrapers)

No.	Condition	Factor
I	**Maintenance**	
	Excellent	1.090
	Average	0.981
	Poor	0.763
II	**Speeds (Maximum)**	
	16 km/h ~ **10 mph**	1.090
	32 km/h ~ **20 mph**	0.872
	48 km/h ~ **30 mph**	0.763
III	**Surface Conditions**	
	Soft Earth — No Rock	1.090
	Soft Earth — Some Rock	0.981
	Well Maintained — Gravel Road	0.981
	Poorly Maintained — Gravel Road	0.763
	Blasted — Sharp Rock	0.654
IV	**Wheel Positions**	
	Trailing	1.090
	Front	0.981
	Driver (Rear Dump)	0.872
	(Bottom Dump)	0.763
	(Self Propelled Scraper)	0.654
V	**Loads** (See No. VIII note)	
	T&RA/ETRTO* Recommended Loading	1.090
	20% Overload	0.872
	40% Overload	0.545
VI	**Curves**	
	None	1.090
	Medium	0.981
	Severe	0.872
VII	**Grades** (Drive Tires Only)	
	Level	1.090
	5% Max.	0.981
	15% Max.	0.763
VIII	**Other Miscellaneous Combinations** (See note below)	
	None	1.090
	Medium	0.981
	Severe	0.872
	Condition VIII is to be used when overloading is present in combination with one or more of the primary conditions of maintenance, speeds, surface conditions and curves. The combination of severe levels in these conditions, together with an overload, will create a new and more serious condition which will contribute to early tire failure to a larger extent than will the individual factors of each condition.	

*Tire and Rim Association/European Tire and Rim Technical Organization.

Type of Tire	Base Average Life Hours	km	Miles
E-3 Std. Bias Tread	2510	40 400	**25,100**
E-4 Bias Xtra Tread	3510	56 500	**35,100**
E-4 Radial Xtra Tread	4200	67 600	**42,000**

Using Base Hours (or Miles), multiply by the appropriate factor for *each* condition to obtain approximate estimated hours (or miles) as the final product.

Example: An off-highway truck equipped with E-4 drive tires running on a well maintained haul road having easy curves and minimum grades and receiving "average" tire maintenance attention but being 20% overloaded:

Condition: I II III IV V VI VII VIII
Factor: 0.981 × 0.872 × 0.981 × 0.872 × 0.872 × 0.981 × 0.981 × 0.981 × 3510 base hours = 2114 hours (say 2100 hours)